

MATH1081 Topic 4: Counting and recursion

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Proposition 4.1 (The addition principle)

Let $A_1, \dots, A_k$ mutually disjoint finite sets, i.e. $A_i \cap A_j = \emptyset$ for all $i \neq j$. Then the cardinality of their union is given by

$$|A_1 \cup \dots \cup A_k| = |A_1| + \dots + |A_k|.$$

Remark 4.2

The hypothesis that the sets are mutually disjoint is crucial.

Proposition 4.3 (The multiplication principle)

Given finite sets $A_1, \dots, A_k$ the cardinality of their Cartesian product is given by

$$|A_1 \times \dots \times A_k| = |A_1| \times \dots \times |A_k|.$$

Theorem 4.4 (Ordered selection with repetition)

Given two finite sets A and B whose cardinalities are a and b respectively. Then the number of functions $f: A \rightarrow B$ is given by b^a , i.e.

$$|\text{Func}(A, B)| = b^a.$$

Theorem 4.5 (Re-arranging distinct objects)

Given a finite set N whose cardinality is n then the number of bijective maps $f: N \rightarrow N$ is given by $n! := n(n-1) \cdots 1$.

Theorem 4.6 (Ordered selection without repetition)

Let R and N be finite sets whose cardinalities are r and n respectively. Then the number of injective functions $f: R \rightarrow N$ is given by

$$P(n, r) := n(n-1) \cdots (n-(r-1)) = \frac{n!}{(n-r)!}.$$

Here we make the convention that $0! = 1$.

Counting up to symmetry

Example 4.7

How many 'words' can be made by rearranging the letters in the word *CHEESE*? If we artificially view

$$C H E E S E \text{ as } C H E_1 E_2 S E_3,$$

we have 5! such words (Theorem 4.5). However for us

$$\begin{aligned} C H E_1 E_2 S E_3 &\sim C H E_1 E_3 S E_2 \sim C H E_2 E_1 S E_3 \\ &\sim C H E_2 E_3 S E_1 \sim C H E_3 E_1 S E_2 \sim C H E_3 E_2 S E_1. \end{aligned}$$

Note:

- The relation $\sim$ on these artificial words is an equivalence relation.
- The elements of any given equivalence class are in one-to-one correspondence with bijective function from $\{E_1, E_2, E_3\} \rightarrow \{E_1, E_2, E_3\}$. That is by Theorem 4.5, each equivalence class is of size 3!.

Therefore the number of words given by rearranging the letters in *CHEESE* is $\frac{5!}{3!}$.

Definition 4.8

For n and r integers for which $0 \leq r \leq n$ define the *binomial coefficients*

$$C(n, r) = \binom{n}{r} := \frac{n!}{(n-r)!r!}.$$

Theorem 4.9 (Unordered selection without repetition)

Given a set N with $|N| = n$, the number of subsets $R \subset N$ so that $|R| = r$ is given by $C(n, r)$.

Example 4.10 (Counting cards)

It consists of **52 cards** comprising:

- **thirteen values** ace **A**, two **2**, three **3**, ..., ten **10**, jack **J**, queen **Q** and king **K**
- in each of **four suits** spades ♠, hearts ♥, diamonds ♦ and clubs ♣.
- The spades and clubs are coloured **black**, while the hearts and diamonds are **red**.

A selection of cards from a pack is referred to as a *hand*. We assume that a hand **may not contain the same card twice**, and that the order of cards in a hand is not important.

Corollary 4.11 (Binomial theorem)

Let n be a non-negative integer the

$$(x + y)^n = \sum_{r=0}^n C(n, r) x^r y^{n-r} = \sum_{r=0}^n \binom{n}{r} x^r y^{n-r}.$$

Proposition 4.12 (Pascal's triangle)

Let r and n be integers so that $0 \leq r \leq n + 1$. Then

$$\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}.$$

We display the binomial coefficients in a table so that row n contains $\binom{n}{r}$:

$n = 0:$						1				
$n = 1:$					1		1			
$n = 2:$				1		2		1		
$n = 3:$			1		3		3		1	
$n = 4:$	1		4		6		4		1	

Theorem 4.13

Let $r \leq n$ be positive integers. Then the number of positive (0 not allowed) integer solutions to

$$x_1 + x_2 + \cdots + x_r = n$$

is given by $\binom{n-1}{r-1}$.

Proof.

Write n using n 1s and $(n-1)$ +s:

$$n = (1 + 1 + 1 + \cdots + 1).$$

To decompose n into r summands we may choose $r-1$ +s and replace them with “) + (“.



Corollary 4.14

Let $r \leq n$ be positive integers. Then the number of non-negative integer (0 is allowed) solutions to

$$y_1 + y_2 + \cdots + y_r = n$$

is given by $\binom{n+r-1}{r-1}$.

Theorem 4.15 (Inclusion-exclusion principle)

Let $A_1, \dots, A_n$ be a finite list of finite sets. Define

$$S_0 = |\emptyset| = 0,$$

$$S_1 = \sum_i |A_i|,$$

$$S_2 = \sum_{\{i,j\} \subset \{1,\dots,n\}} |A_i \cap A_j|,$$

$$S_3 = \sum_{\{i,j,k\} \subset \{1,\dots,n\}} |A_i \cap A_j \cap A_k|,$$

$\vdots$

$$S_n = |A_1 \cap \dots \cap A_n|.$$

Then

$$|A_1 \cup \dots \cup A_n| = \sum_{r=0}^n (-1)^{r+1} S_r.$$

The Pigeonhole Principle

Gently place $n + 1$ pigeons into n pigeonholes then is a least one pigeonhole with two or more pigeons

Theorem 4.16

Let A be a set of size r and $A_1, \dots, A_n$ be n mutually disjoint subsets of A such that $A = A_1 \cup \dots \cup A_n$. If $r > n$ then there exists $1 \leq i \leq n$ such that $|A_i| > 1$.

Corollary 4.17 (Pigeonhole principle)

Let $f: A \rightarrow B$ be a function from a set A of size r to a set B of size n . If $r > n$ then f is not injective.

Corollary 4.18 (Generalised pigeonhole principle)

Let $f: A \rightarrow B$ be a function from a set A of size r to a set B of size n . Then there is some $b \in B$ so that

$$f^{-1}(b) := |\{a \in A \mid f(a) = b\}| = \left\lceil \frac{r}{n} \right\rceil.$$

4.3 Recurrence

We may define a sequence of natural numbers inductively using a *recurrence relation*.

Example 4.19 (Fibonacci numbers)

The *Fibonacci numbers* F_n are defined by the following recurrence relation

$$F_n = F_{n-1} + F_{n-2},$$

with the *initial conditions* $F_0 = 0$ and $F_1 = 1$.

Definition 4.20

An equation giving the value of a sequence a_n in terms of

$$a_{n-1}, a_{n-2}, \dots, a_{n-k}$$

is called a *recurrence relation of order k* .

To determine the value of a_n one needs more information in the shape of *initial conditions*. For an order k recurrence relation these are the first k values values, i.e. the values of $a_0, \dots, a_{k-1}$.

Notation 4.21

A general recurrence relation of order k looks like this

$$a_n + c_1 a_{n-1} + \cdots + c_k a_{n-k} = f(n)$$

here if the coefficients c_i can be an arbitrary function.

- The recurrence relation is **linear** if c_i are constants.
- The recurrence relation is **homogeneous** if $f(n) = 0$.
- Given sufficient initial conditions we say a recurrence has a **solution** if a_n can be written explicitly in terms of n only.

Theorem 4.22

The linear homogeneous recurrence of order 1

$$a_n = ra_{n-1} \text{ with } a_0 = A$$

has a solution $a_n = Ar^n$.

Theorem 4.23

Consider the second order linear homogeneous recurrence

$$a_n + p a_{n-1} + q a_{n-2} = 0;$$

let the solutions to $r^2 + pr + q = 0$ be α and β .

- If α, β are real and $\alpha \neq \beta$ then the general solution is

$$a_n = A\alpha^n + B\beta^n.$$

- If $\alpha = \beta$ then the general solution is

$$a_n = A\alpha^n + Bn\beta^n.$$

- If α, β are complex conjugates $\rho e^{\pm i\theta}$ then the general solution is

$$a_n = \rho^n (A \cos(n\theta) + B \sin(n\theta)).$$

Here A and B are some constants.

Notation 4.24

The polynomial $r^2 + pr + q$ in Theorem 4.23 is called the *characteristic polynomial*.

Remark 4.25

The same technique can be applied to linear homogeneous recurrences of arbitrary order. Compute roots of the characteristic polynomial...

Theorem 4.26

The general solution of a non-homogeneous linear second order recurrence

$$a_n + pa_{n-1} + qa_{n-2} = f(n) \quad (1)$$

is of the form

$$a_n = h_n + p_n.$$

Here h_n is the general solution of the homogeneous equation

$$a_n + pa_{n-1} + qa_{n-2} = 0,$$

and p_n is some 'particular' solution to the non-homogeneous equation (1).

Proposition 4.27 (How does one guess?)

This is not pretty unfortunately. The point is to mirror the form of $f(n)$.

<i>Form of $f(n)$</i>	<i>Guess for particular solution</i>
<i>(a) Constant</i>	$p_n = c$
<i>(b) Polynomial of degree s in n</i>	$p_n = b_s n^s + \cdots b_1 n + b_0$
<i>(c) Exponential form α^n</i>	$p_n = c\alpha^n$
<i>(d) Product of two previous forms</i>	p_n product of corresponding guesses

If the guess for p_n doesn't work then multiply the guess in the table by n ; and multiply by another factor of n or more if necessary.